The 99-Cent Question: How Accurate Are Kalshi’s Odds?

What 7 million hourly prices say about Kalshi, and whether 90–99¢ favorites are as safe as they look

STATS 414 MIDTERM · OCTOBER 28, 2026

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ROADMAP

Three questions about Kalshi prices

1

What does Kalshi trade?

Markets, volume and liquidity across 840,000 markets

2

Are its prices right?

Calibration across 1–99¢, then a zoom on 90–99¢ favorites

3

Can a model beat the price?

Logistic, random forest and XGBoost vs the quoted price

Why favorites? A YES contract pays $1 or $0, so its price is the crowd's probability. Buying at 99¢ risks 99¢ to make 1¢: one loss wipes out 99 wins. Whether that is a good bet depends entirely on how accurate the price is.

DATA

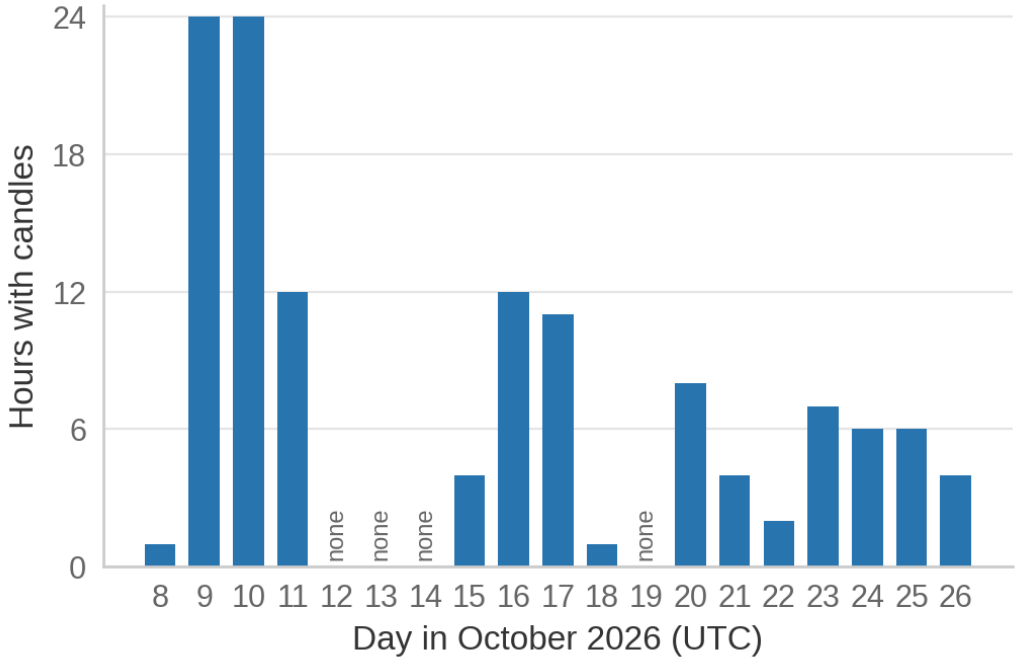
Every hourly Kalshi price for three weeks, with gaps

7.17M

hourly candles (bid, ask, volume), Oct 8–27

839,613

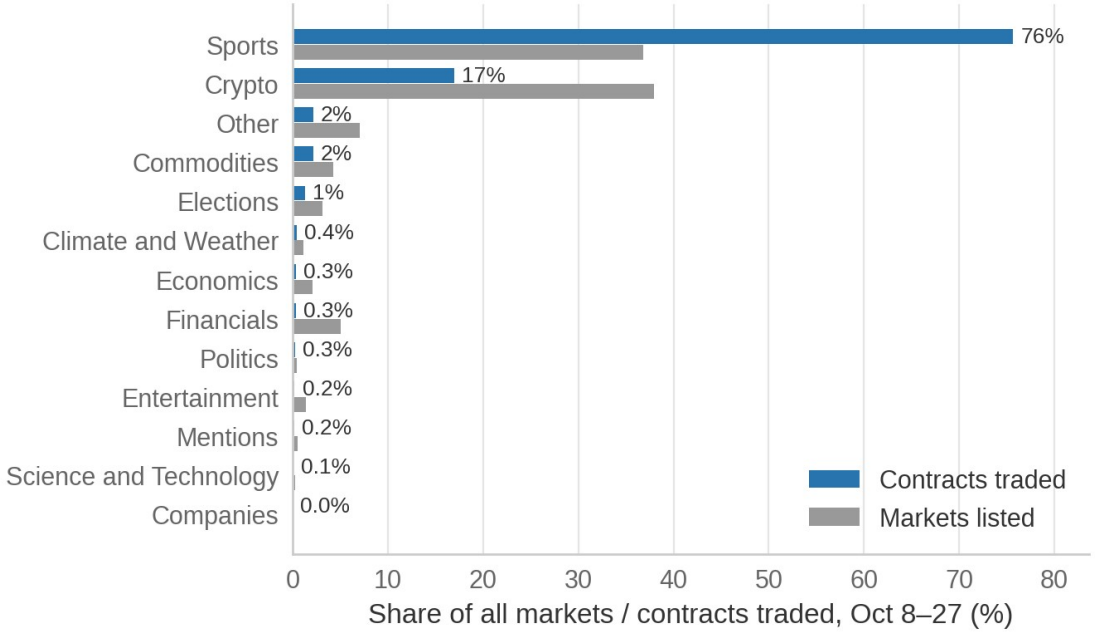
markets in 67,261 events



Data: Kalshi course database, hourly candles Oct 8–27, 2026 · hours with at least one candle per UTC day

EXPLORATION 1

Sports and crypto are most of Kalshi



76%

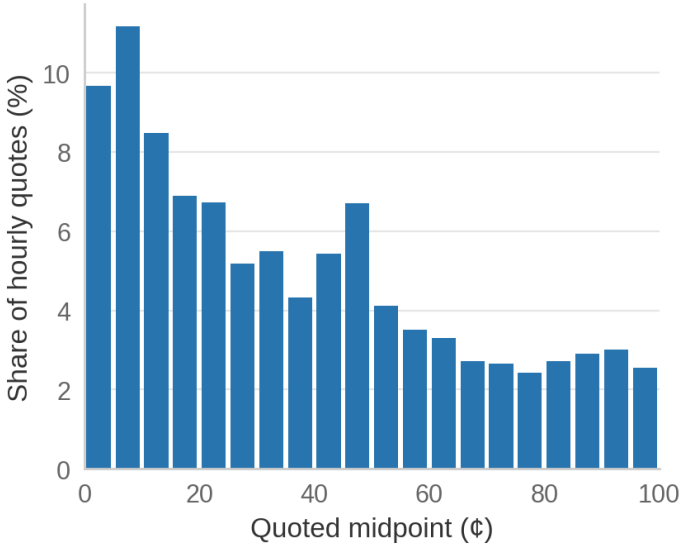
of contracts traded were Sports, from 37% of markets

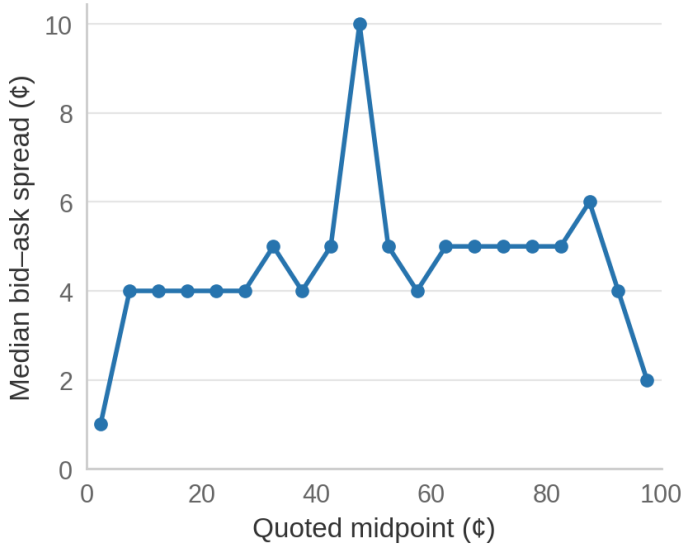
- The top 1% of markets carry 82% of all volume
- 72% of markets never traded in the window
- Politics, Economics, Tech and Entertainment: each under 0.4% of volume

Data: Kalshi course database, hourly candles Oct 8–27, 2026 · volume = contracts traded, summed over hourly candles

EXPLORATION 2

Most quotes are long shots; spreads peak at 50¢



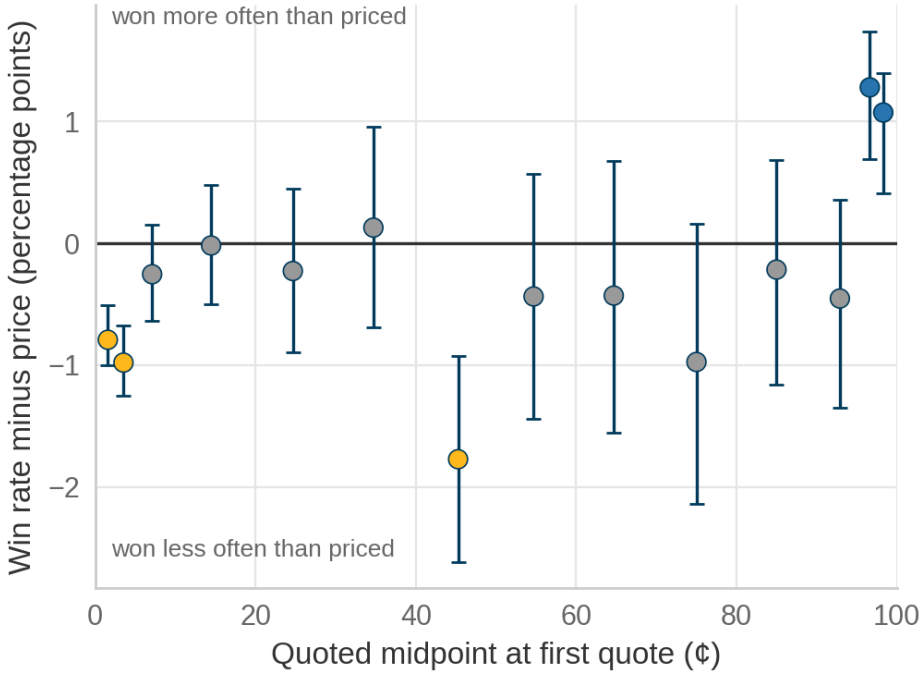


- Left: 21% of hourly quotes are below 10¢ and 5.5% above 90¢, because events list one market per outcome
- Right: the median spread is 4¢ overall, 1–2¢ at the extremes and widest near 50¢, where outcomes are least certain

Data: Kalshi course database, hourly candles Oct 8–27, 2026 · 5.5M valid two-sided quotes (0 < bid ≤ ask < 1)

EXPLORATION 3

Prices are calibrated except at the extremes



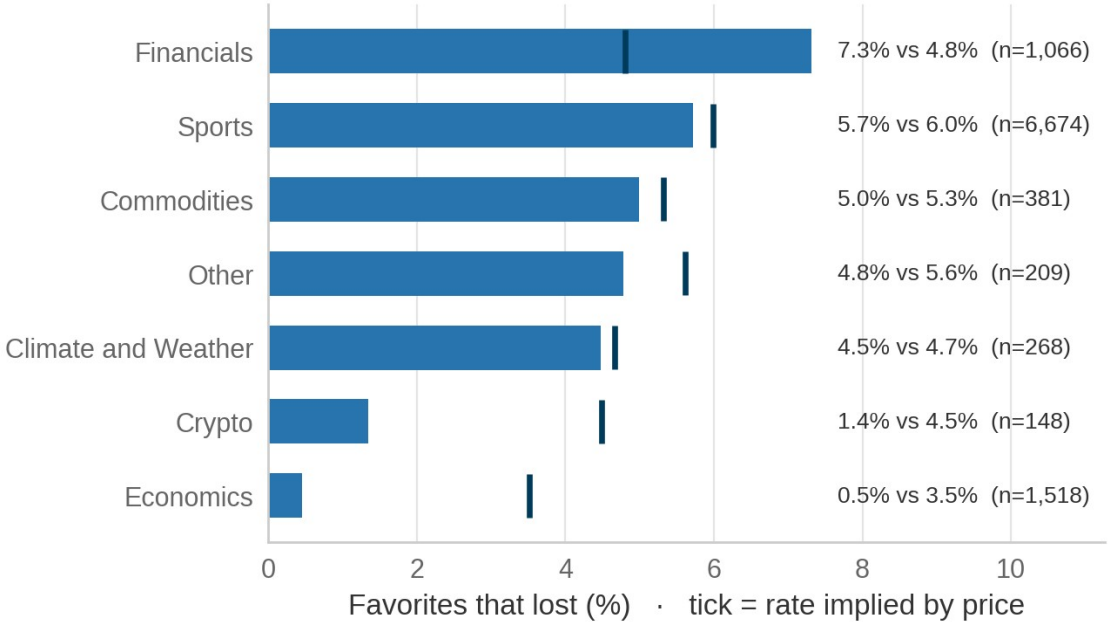
0.5 pts

average gap between win rate and price, 129,866 markets

- Long shots under 2¢ won 0.8% of the time vs 1.6% priced; 2–5¢ won 2.5% vs 3.5%
- 95–100¢ favorites won 1 point more often than priced
- Gold: long shots overpriced; blue: favorites underpriced

Data: Kalshi course database, hourly candles Oct 8–27, 2026 · first quote per finalized market (spread ≤ 5¢) · 95% Wilson intervals

Economics favorites were underpriced



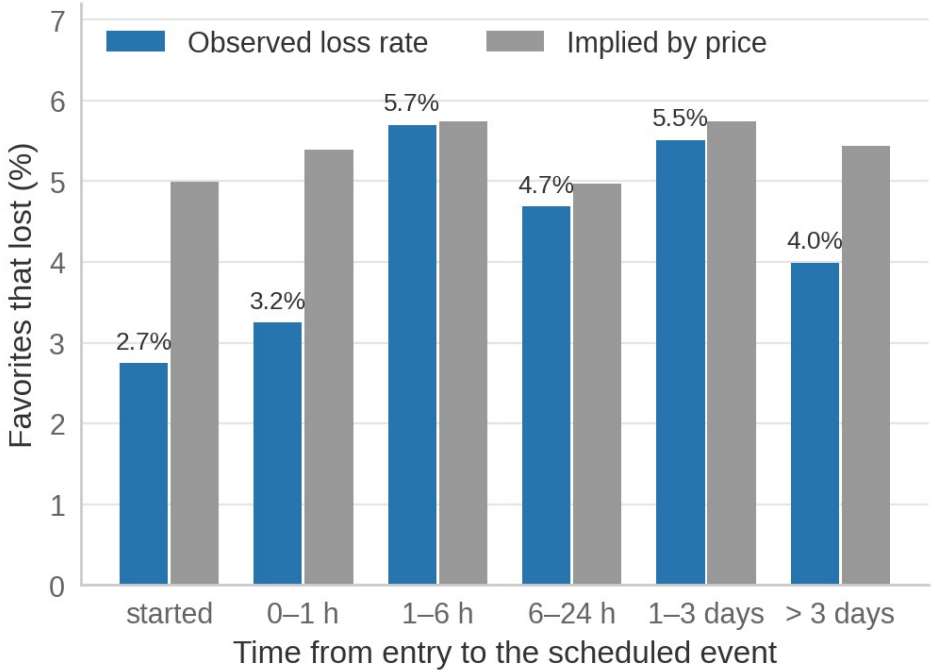
10,351 favorites, 513 lost
First quote per market with ask 90–99¢, midpoint ≥ 90¢, spread ≤ 5¢

Economics
0.5% lost vs 3.5% implied; 95% CI of the gap [−3.4, −2.6] pts; held in every period; mostly AAA gas-price markets

Financials 7.3% vs 4.8%, but its CI includes 0

Data: Kalshi course database, hourly candles Oct 8–27, 2026 · intervals resample whole events (1,000 bootstrap draws)

Favorites lose as priced until the event starts



- From 1 hour to 3 days out, loss rates match the price
- At or after the scheduled event start: 2.7–3.2% lost vs about 5% implied
- A small effect, and not stable across periods

Data: Kalshi course database, hourly candles Oct 8–27, 2026 · time from entry to the scheduled event time · all periods, n = 10,351

MODELS

We split by date and tuned three model families

Train

Entries Oct 9–11

4,358 markets · 179 losses

Validation

Entries Oct 15–18

3,694 markets · 220 losses

Test

Entries Oct 20–25

2,299 markets · 114 losses

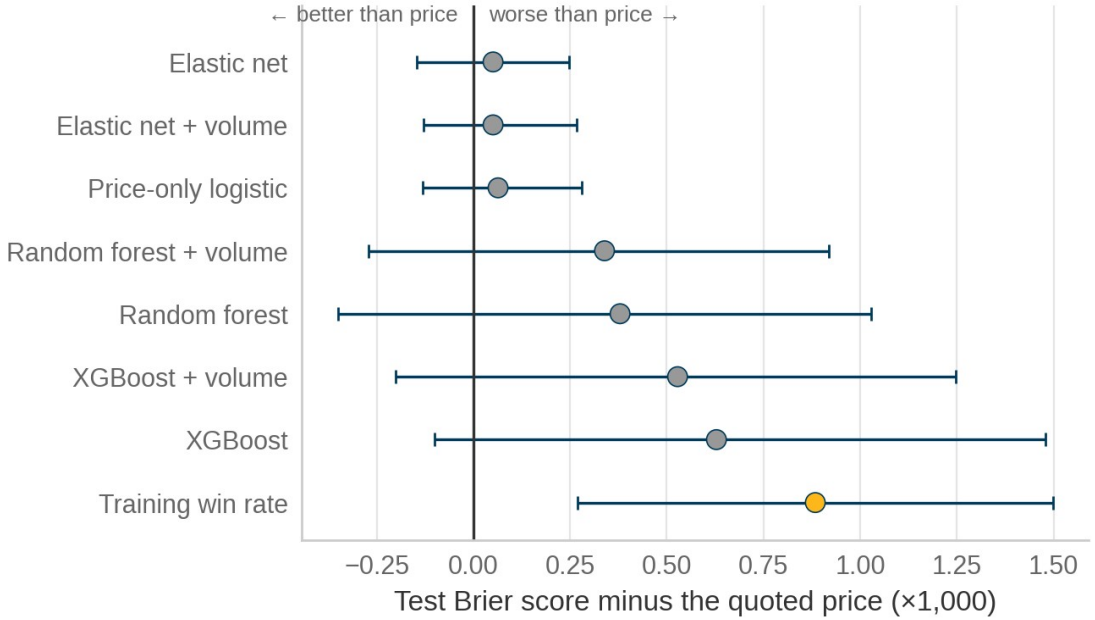
Features at entry

- Price, spread, open interest
- Volume now and in the prior 6 h
- Hours to the scheduled event; recent price change
- Category; markets in the event

Models (tuned on validation Brier)

- Elastic-net logistic (Week 3): C = 0.01, L1 share 0.5
- Random forest (Week 4): depth 4
- XGBoost (Week 4): depth 2, learning rate 0.03
- Each with and without volume; baseline = the quoted price

No model beat the quoted price



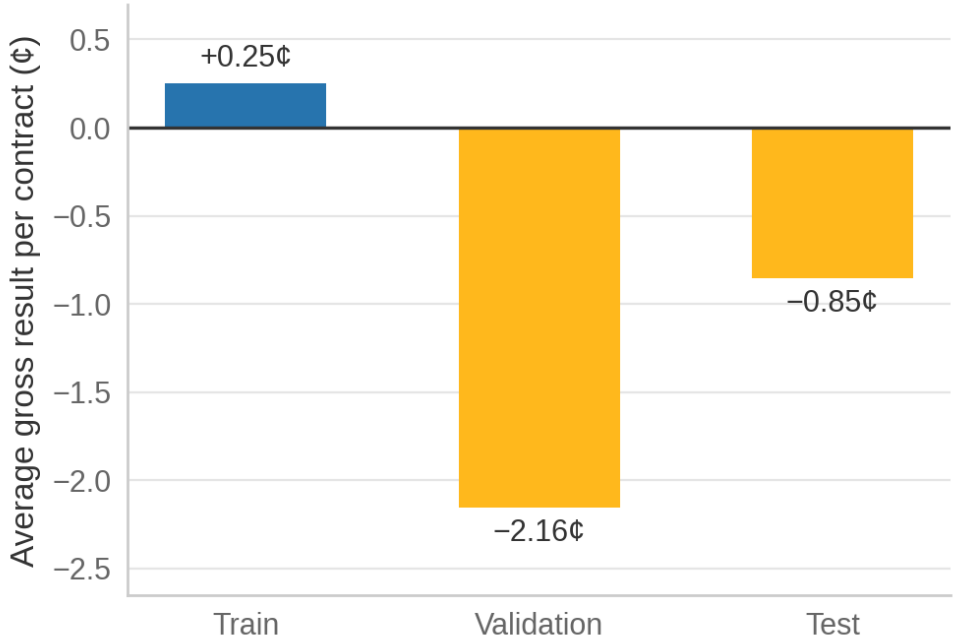
0.04624

test Brier score of the quoted price: the best of all

- Closest: elastic net, 0.04629 (gap CI −0.00015 to +0.00025)
- Volume changed Brier by at most 0.0001 in every family
- The elastic net set every volume, timing and category coefficient to 0

Data: Kalshi course database, Oct 8–27, 2026 · test n = 2,299 · event-bootstrap 95% CIs · gray: CI includes 0, gold: worse

Buying every favorite at the ask still lost money



−$19.61

on 2,299 test contracts, even though 95.0% of them won

- Per contract: −0.85¢ in test, −2.16¢ in validation, +0.25¢ in train
- The small favorite edge from slide 6 does not survive buying at the ask

Data: Kalshi course database, Oct 8–27, 2026 · one contract at the displayed ask, held to settlement, before fees

What we learned

- Kalshi is mostly sports and crypto, and most markets barely trade
- Prices are calibrated within half a point, with a favorite–long-shot bias at the ends
- No model beat the price for favorites, and buying them at the ask lost money

Thank you

Questions?

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